

Smriti Singh

9471723727 | smriti.shaurya22@gmail.com | linkedin.com/in/smriti-singh-83541b267 | github.com/SmrSingh

SUMMARY

AI/ML Engineer with expertise in deep learning, predictive modeling, and tabular data classification. Experienced in developing and integrating ML models through academic projects in drug discovery, crop recommendation, and power consumption analysis. Proficient in Python, TensorFlow, PyTorch, and C++, with working knowledge of Java. Strong in problem-solving, research, collaboration, leadership, and project planning. Actively seeking a dynamic developer role to apply skills and build impactful AI-driven solutions.

TECHNICAL SKILLS

Programming Languages: Python, C++, SQL, Java

Deep Learning Frameworks: Predictive Modeling, Statistical Analysis, Algorithms

Libraries & Tools: NumPy, Pandas, Scikit-learn, OpenCV, NLTK, Git, Docker

PROJECTS

Power Consumption Analysis

May 2025 – July 2025

Machine Learning Project

Python, Scikit-learn, Pandas, TensorFlow

- Built an ML system to analyze household power usage, achieving 86.2% accuracy in identifying peak consumption and inefficient appliances.
- Integrated weather data and time-of-use pricing for personalized energy-saving recommendations.
- Designed interactive dashboards to visualize trends and cost-saving opportunities.

Crop Recommendation System

Feb 2025 – April 2025

Full-Stack ML Project

Python, Flask, MongoDB, JavaScript

- Designed and developed an end-to-end crop recommendation system leveraging geolocation and environmental factors.
- Built and deployed machine learning models to analyze soil nutrients (N, P, K), rainfall, and climate data for optimal predictions.
- Developed RESTful APIs using Flask for prediction services and Node.js for real-time environmental data handlings.
- Created an interactive frontend with OpenWeather API integration, enabling dynamic, location-based crop recommendations.

Drug Discovery Model

Aug 2024 – Oct 2024

ML Research Project

Python, TensorFlow, PyTorch, RDKit

- Developed a predictive machine learning model leveraging Graph Neural Networks (GNNs) to accelerate drug discovery by analyzing molecular bioactivity.
- Engineered molecular graph representations (SMILES → graph structures) to effectively capture structural relationships for target interaction prediction.
- Achieved 87.9 accuracy in bioactivity classification, significantly outperforming baseline models.

EDUCATION

VIT Bhopal University, Bhopal

Sep 2023 – Present

Bachelor of Technology in Computer Science (AI & ML Specialization)
CGPA: 9.14

Vidya Bharati Chinmaya Vidyalaya, Jamshedpur

2020 – 2022

12th Grade: 86.8% (2022)

Hill Top School, Jamshedpur

2014 – 2020

10th Grade: 94% (2020)

CERTIFICATIONS

- Microsoft Azure Data Fundamentals Certified.
- Machine Learning in Collaboration with Google for Developers.